

Web Data Integration

Introduction to the Student Projects



Agenda

1. Overview
 - Phase I: Data Selection and Data Translation
 - Phase II: Identity Resolution
 - Phase III: Data Fusion
2. Details about Phase I: Data Collection and Data Translation
 - Requirements
 - Examples
 - Data Sources
3. Group Formation
4. Start of Group Work

Overview Student Projects

- Phase I: Data Selection and Data Translation

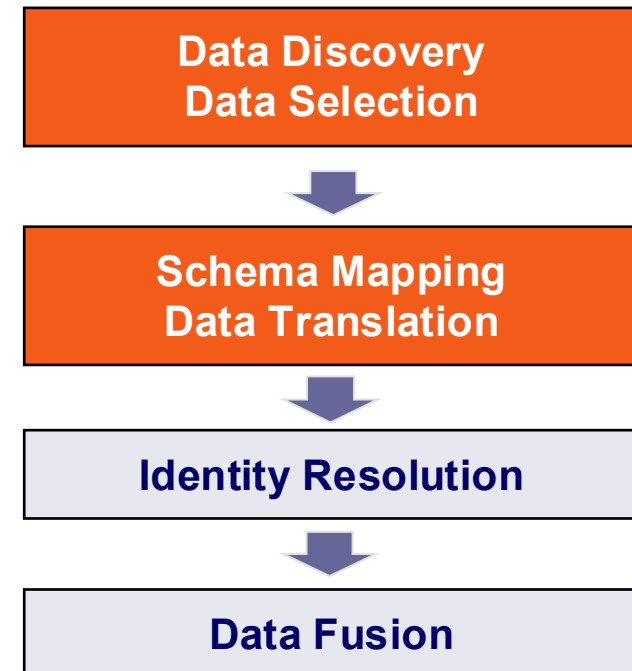
Duration: now till October 29th

Tasks:

1. Find a partner (groups of five)
2. Decide on a use case
3. Collect data from the Web
4. Profile your data and write outline about profile
5. Generate integrated schema (target schema)
6. Convert all your data into the integrated schema using MapForce

Result: All data is represented using a single unified schema

- one JSON or XML file per data source



Overview Student Projects

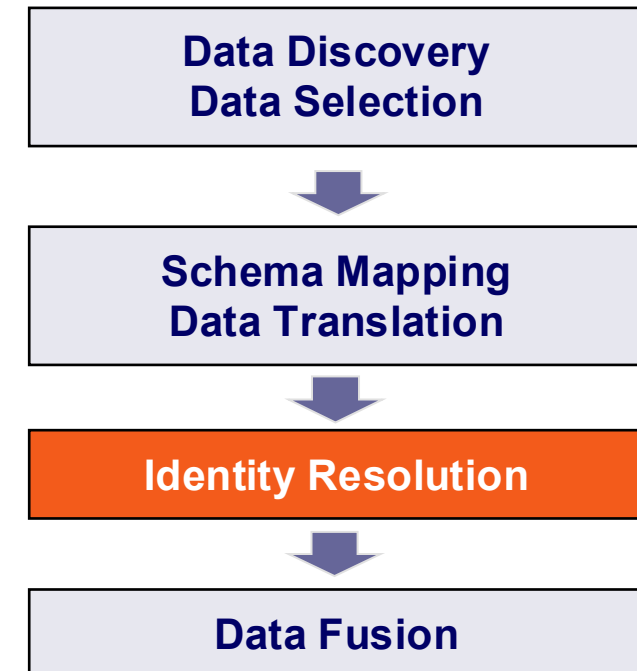
- **Phase II: Identity Resolution**

Duration: October 29th – November 12th

Tasks: Use PyDI framework to

1. Identify records in different data sets that describe the same entity
2. Experiment with different combinations of similarity measures
3. Use blocking to speed up the comparisons
4. Evaluate quality of your approach

Result: Correspondences between records in different data sets that describe the same entity



Overview Student Projects

- **Phase III: Data Fusion**

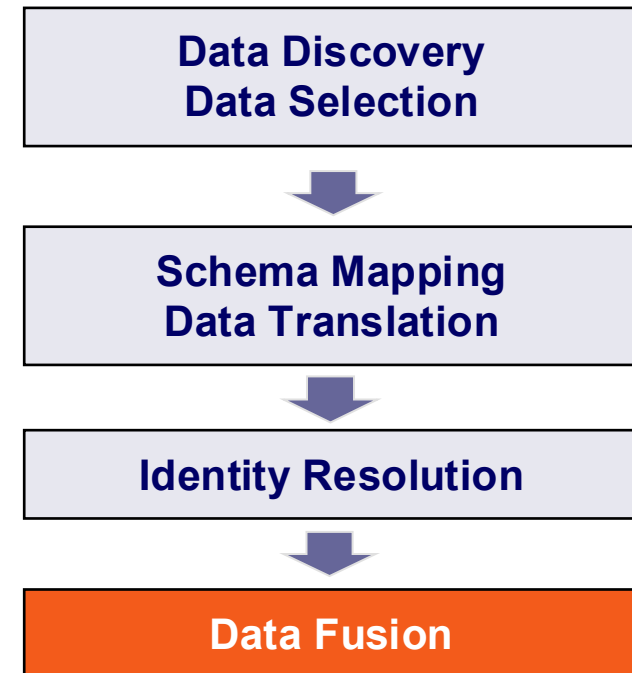
Duration: November 12th – November 29th

Tasks: Use PyDI framework to

1. Merge data and resolve data conflicts
2. Experiment with different conflict resolution strategies
3. Measure the quality and completeness of the final fused data set

Results:

1. Fused data set in which each real-world entity is described by only a single record and these records contain no data conflicts
2. Project report (12 pages) summarizing the results of the phases 1-3



Overview Student Projects

- **Final Presentations**

- Dates: December 2nd and December 3rd
- Overview of your use case
- Explain your data
- Explain the strategies that you used in each step
- Discuss the quality of your solution of each step



Grading of the Projects (IE683, 3 ECTS)

Individual contribution to:

70%: Project work

- quality of your solution
- systematic experimentation with different alternatives
- systematic evaluation of experiments
- quality of written report

30%: Final presentation

- structure
- slides
- discussion

Please submit table on **who did what** together with the report.

Coaching Sessions

- I will give you tips and answer questions concerning your project.
- Registration via **e-mail** is mandatory!
 - until Monday/Tuesday night (min. 2 days before the coaching session)!
 - including the questions that you like to discuss
- You are required to attend at least one coaching session!

Schedule

Week	Wednesday	Thursday
09.09.2026	Lecture: Introduction to Web Data Integration	Lecture: Structured Data on the Web
16.09.2026	Lecture: Data Exchange Formats – Part 1	Lecture: Data Exchange Formats – Part 2
23.09.2026	Exercise: JSON, XML, and Information Extraction	Project: Introduction to Student Projects
30.09.2026	Lecture: Data Discovery	Lecture: Schema Mapping
07.10.2026	Lecture: Schema Matching	Project: Feedback about Project Outlines
14.10.2026	Exercise: Introduction to MapForce	Project Work: Schema Mapping
21.10.2026	Coaching: Schema Mapping	Lecture: Identity Resolution
28.10.2026	Lecture: Identity Resolution	Exercise: Identity Resolution
04.11.2026	Project Work: Identity Resolution	Coaching: Identity Resolution
11.11.2026	Lecture: Data Fusion	Exercise: Data Fusion
18.11.2026	Project Work: Data Fusion	Coaching: Data Fusion
25.11.2026	Project Work: Data Fusion	Project Work: Data Fusion
02.12.2026	Presentation of Project Results	Presentation of Project Results
17.12.2026	Final Exam	

Administrative Requirement – Exam Registration

- In order to receive a grade for your project, you are required to register for the exam through Portal2.
- The registration period is from the 21st of October to the 4th of November.



Details about Phase I: Data Collection and Data Translation

- Duration: now – October 29th
- **Today**
 1. Form teams of **five** people
 2. Decide on a domain/use case
 3. Start data collection and profiling
- **Until Tuesday, October 6th, 23:59**
 - Send a 4 page abstract on your project (details next slide)
- **Thursday, October 8th, 13:45-15:15**
 - You get feedback on your abstract (if necessary)
- **Wednesday, October 14th, 15:30-17:00**
 1. Introduction to **MapForce**
 2. Start using MapForce to translate data to target schema



Project Requirements

You should integrate:

1. **3 different data sets**
2. at least **2,500 entities** described in total (in joint dataset)
 - but more are better, good: >10,000 but <100,000
3. at least **1,000 entities** should be contained in at least **two datasets**
 - please estimate based on small sample
4. at least **8 attributes** in joint dataset
 - entities should be identifiable by attribute combinations of at **least two attributes**, e.g. name+birthdate
5. at least **5 attributes** should be contained in at least **two datasets**
 - some attributes (other than name) should be contained in three datasets (for fusion by voting)
6. ideally, at least one of your attributes is a **list attribute**
 - actors of a movie, directors of a company, songs on a CD

Project Abstracts

- Purpose of project abstract
 - check whether your ideas are feasible
 - proof that you fulfill the requirements (last slide)
- Content
 1. Brief description of use case
 2. Explanation how the datasets fulfill the requirements
 1. Schema and basic profile of each dataset
 - number of records per class
 - attributes with high percentage of missing values
 2. Integrated schema and overlap with input schemata
 3. Explanation why enough entities are likely contained in multiple datasets
- Submit via ILIAS to me
- **Deadline: Tuesday, October 6th, 23:59**



Tables that MUST be used in Project Abstracts

1. Schema and Basic Profile of each Data Set

Table 1. Datasets

Dataset	Source(*)	Format	# of entities	# of attributes	List of attributes (**)
IMDB	Download URL	csv	17,000	10	title, director (MV), year,...
DBpedia	Dbpedia.org/sparql	xml	23,500	8	name, birthDate, activeYears,..
Freebase	Download URL	csv	11,000	14	first name, surname, spouse,...

(*) Should explain where from and how you got the data

(**) Mark attributes with >30% missing values (MV)

2. Integrated Schema and Overlap with Input Schemata

Table 2. Attribute Intersection with Integrated Schema

Attribute name	Attribute type	Datasets in which the attribute is found
name	string	dataset1, dataset2, dataset3 (use proper dataset names)
director	string/list	dataset1, dataset3
year	date	dataset2, dataset3
...		...

Requirements for the Final Project Report

- **12 pages (sharp!)** – counted without title page, table of content, literature list
 - Every extra page (including appendix pages) will reduce your mark by 0.33
- Due to **Sunday, 29th November 2026, 23:59**
 - Submit via ILIAS
- You must use latex template of **Springer CS Proceedings**
 - <http://www.springer.com/de/it-informatik/lncs/conference-proceedings-guidelines>
- Also **submit**
 - your **code** and
 - (a subset) of your **data**
- Please cite sources properly if you use any
 - Preferred citation style [Author, year]

Possible Use Cases for Student Projects

- Movies
 - budget, actors, directors, oscar nominations...
- Video Games
 - name, platform, genre, multi-player, studio, year, budget, sales ...
- Companies
 - sector, performance, key persons, ownership, venture capital
- Songs
 - title, album, artist, releases, genre, producer, composer...
- Musicians
 - first name, last name, birth date, birth place, bands, albums ...
- Products
 - name, model number, weight, height, width, memory, product category

Example Use Case 1: Movies

- Individual Data Sets contain
 - Movies
 - Actors
 - Directors
 - Oscar Nominations & Wins
 - Golden Globe Nominations & Wins
- Integrated dataset will contain
 1. Movies with release date, budget,... and awards nominated/won
 2. lists of actors and directors per movie



Example Use Case 1: Movies

- Lists of Oscar/Golden Globe nominees and winners
 - <https://www.kaggle.com/datasets/unanimad/the-oscar-award>
 - <https://www.kaggle.com/datasets/dimitarmitkov/golden-globe-awards-2024>
- List of The Guardian greatest films (by Genre)
 - <https://www.theguardian.com/news/datablog/2010/oct/16/greatest-films-of-all-time>
- A large movie list
 - <https://www.kaggle.com/datasets/chaitanyahivlekar/large-movie-dataset>



Example Use Case 1: Movies



- Movie data from **DBpedia**
- Issue a SPARQL query against <http://dbpedia.org/sparql>
- Result can be stored as CSV, JSON, XML, ...

```
SELECT ?title ?budget ?gross ?director
WHERE {
    ?x a dbo:Film .
        ?x dbo:budget ?budget .
        ?x dbo:gross ?gross .
    ?x dbo:director ?d .
        ?d foaf:name ?director .
        ?x rdfs:label ?title .
    FILTER(LANG(?title)="en")
}
```

Example Use Case 2: Companies

- Goal: Combine multiple datasets into a single dataset having the following attributes:
 - company name
 - website
 - founding date
 - headquarters country (regional branch vs. company)
 - headquarters city
 - industry (single taxonomy)
 - assets (normalized)
 - revenue (normalized)
 - key persons (list)
 - founders (list)

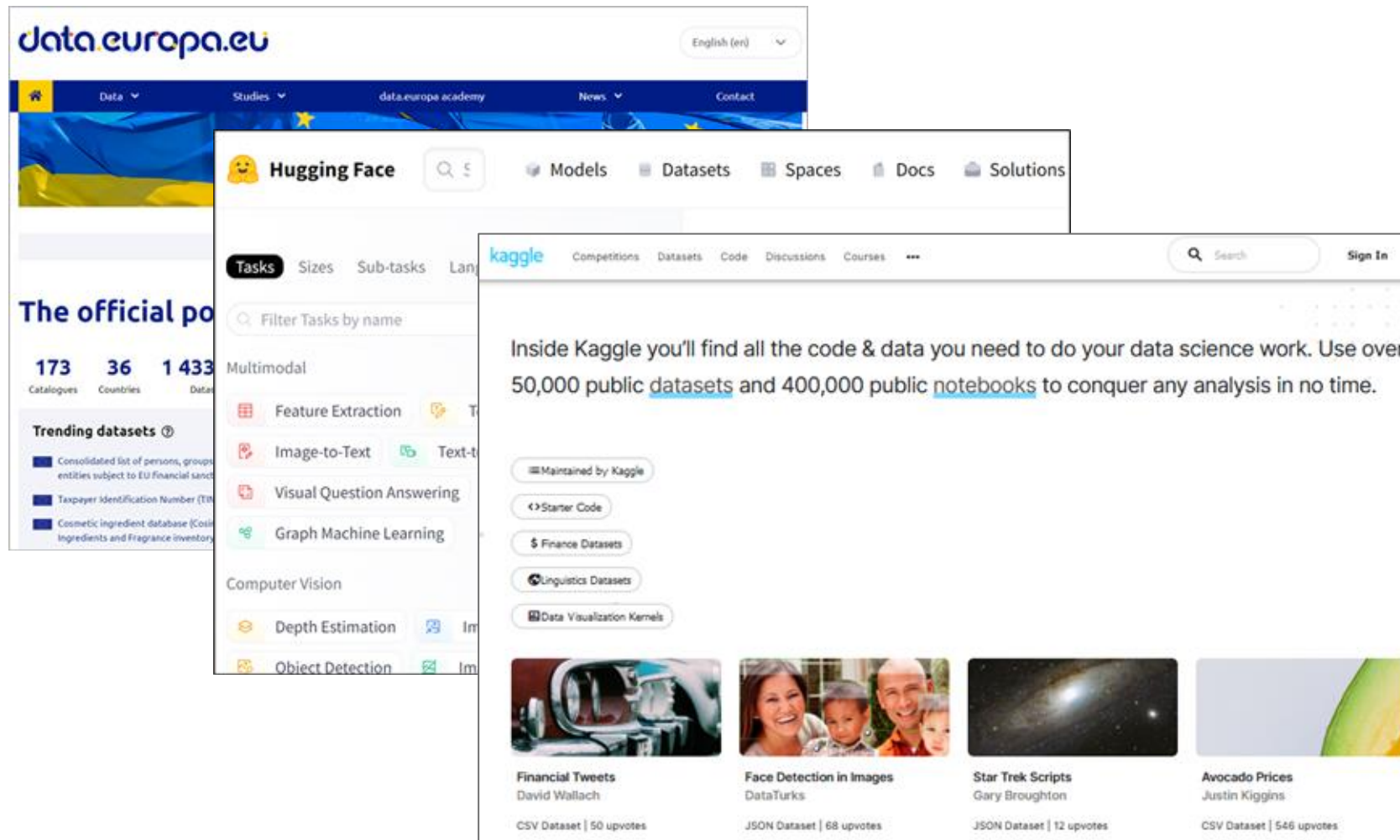
Example: Data Sources about Companies

- Forbes data set with top 2000 companies worldwide
 - <https://www.kaggle.com/ash316/forbes-top-2000-companies>
- Open Data 500 Companies data set with 500 US located companies
 - <https://www.kaggle.com/govlab/open-data-500-companies>
- Kaggle data set with 7.1M companies
 - <https://www.kaggle.com/kaleab1/companies>
- Company data from DBpedia

```
SELECT ?name ?ind_label ?equity ?income
WHERE {
  ?x a dbo:Company .
      ?x rdfs:label ?name .
      ?x dbo:industry ?industry .
      ?industry rdfs:label ?ind_label .
      ?x dbo:equity ?equity .
      ?x dbo:netIncome ?income .
      FILTER (LANG(?name)="en"
              && LANG(? ind_label)="en")}
```

Where do I find Data for my Project?

- Data Portals



Where do I find Data for my Project?

- Google Dataset Search
 - <https://toolbox.google.com/datasetsearch>



The image displays two screenshots of the Google Dataset Search interface. The left screenshot shows search results for the query "Company listing". The right screenshot shows search results for the query "Cities".

Left Screenshot (Query: Company listing):

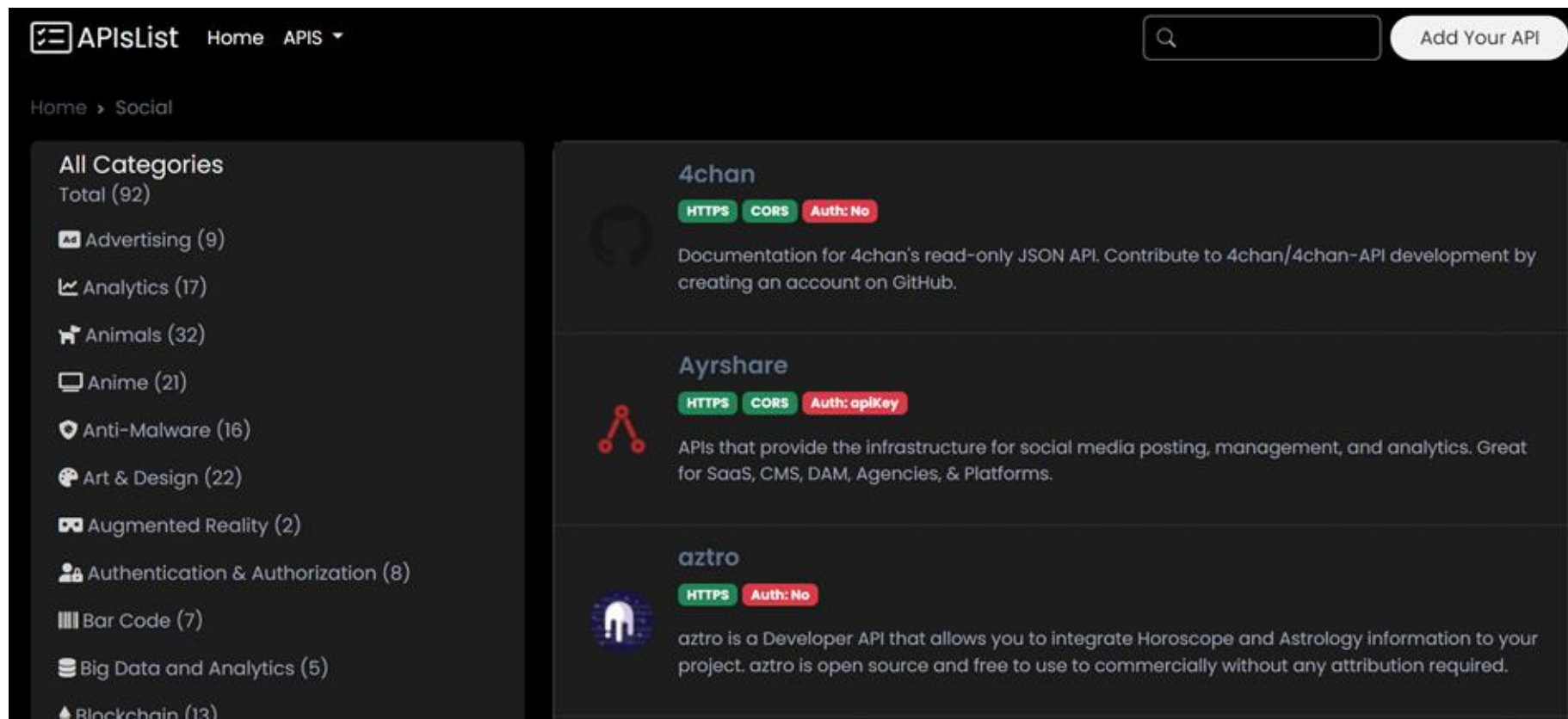
- Business Registration Data** (data.ct.gov, Updated Nov 20, 2015)
- Italy IT: No of Listed Domestic Companies: Total** (www.ceicdata.com)
- EDGAR Database of SEC Filings** (data.wu.ac.at)

Right Screenshot (Query: Cities):

- Cities and Towns NTAD** (data.world, Updated Sep 11, 2018)
- World Cities** (hub.arcgis.com, Published Jun 30, 2013)
- World Cities** (www.kaggle.com, Updated Jun 13, 2017)

Where do I find Data for my Project?

- **Web APIs**
 - e.g., apislist.com – lists many public APIs
 - requires some additional effort (using the API and getting the data)



Where do I find Data for my Project?

- DBpedia, Wikidata and other Linked Data sources
- Look at a single resource
 - http://dbpedia.org/resource/2001:_A_Space_Odyssey_%28film%29
- Look which properties are there (preferable dbpedia-owl)
- Construct a SPARQL query
- Go to <http://dbpedia.org/sparql> and get the data
- Hint: use **OPTIONAL** for properties that are not present for all entities:

```
SELECT ?title ?budget ?gross ?director
WHERE {
    ?x a dbo:Film .
    OPTIONAL {?x dbo:gross ?gross . }
    ...
}
```

There are 87,000 Films
in DBpedia, but only
9,000 with gross



Where do I find Data for my Project?



- **Schema.org data** from multiple web sites
 - Product, local business, hotel, job posting,
- <http://webdatacommons.org/structureddata/schemaorgtables/>
- <http://www.webdatacommons.org/structureddata/>

Class-Specific Subsets of the Schema.org Data

Class Name	Total Number of	Top Classes (Entity Count)	Total File Size	Quad File
http://schema.org/AdministrativeArea	Quads: 1,724,857 URLs: 85,625 Hosts: 63	http://schema.org/AdministrativeArea (100,671) http://schema.org/GeoCoordinates (84,152) http://schema.org/Country (83,851) http://schema.org/Continent (83,567)	23 MB	schemaorgAdministrativeArea.nq.gz (sample)
http://schema.org/Airport	Quads: 80,258,863 URLs: 963,538 Hosts: 99	http://schema.org/Airport (26,764,415) http://schema.org/PostalAddress (9,238) http://schema.org/Product (1,290) http://schema.org/Offer (1,283)	961 MB	schemaorgAirport.nq.gz (sample)
http://schema.org/PostalAddress	Quads: 776,573,609 URLs: 13,475,055 Hosts: 131,064	http://schema.org/PostalAddress (48,086,763) http://schema.org/LocalBusiness (16,641,260) http://schema.org/GeoCoordinates (12,345,942) http://schema.org/Place (9,071,774)	14,364 MB	schemaorgPostalAddress.nq.gz (sample)
http://schema.org/Product	Quads: 2,829,523,589 URLs: 48,314,143 Hosts: 104,118	http://schema.org/Product (287,815,069) http://schema.org/Offer (221,781,710) http://schema.org/AggregateRating (38,398,548) http://schema.org/Review (26,209,678)	62,179 MB	schemaorgProduct.nq.gz (sample)

Creating an Integrated Schema

1. Have a look at your input data
 - Which entities exist? What attributes do they have?
 2. Check input data against project requirements (see **Slide 12**)
 - Create the tables for the project abstract (see **Slide 14**)
 3. Apply schema integration method from lecture
 - Rules of Thumb or Spaccapietra, et al.
- E.g.
- Movie: title, date, budget, revenue, Oscar, director, actors (list)

Creating an Integrated Schema

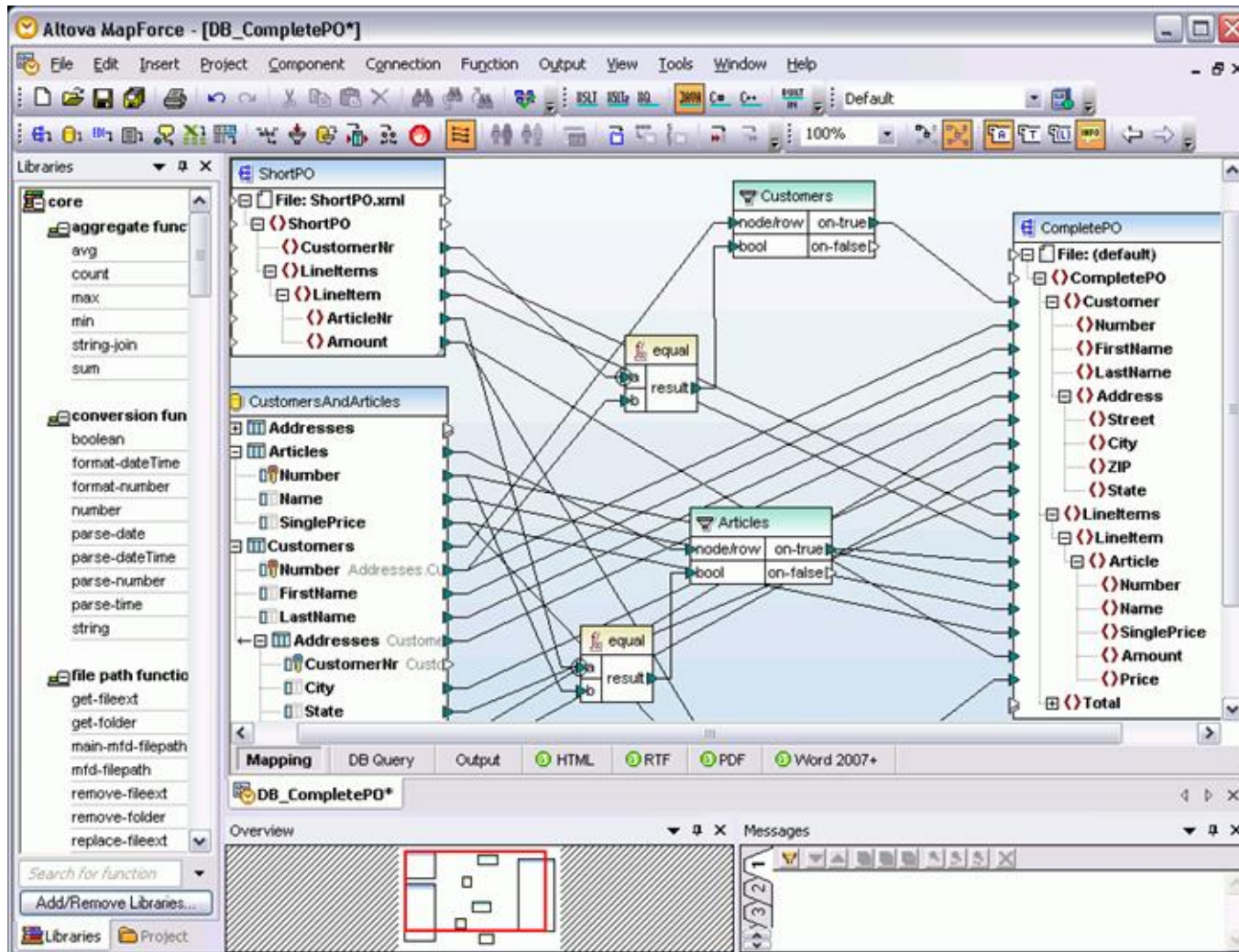
Hint: Create an example XML file

- using the integrated schema
- for some data from each input source
- in order to check if integrated schema can represent input data.

```
<movies>
  <movie>
    <title>2001</title>
    <director>
      <firstname>Stanley</firstname>
      <lastname>Kubrick</lastname>
      ...
    </director>
  </movie>
  ...
</movies>
```

Outlook: Exercise 14.10

1. Introduction to MapForce
2. Start translating your data into the unifying schema using MapForce



...and now

1. Team formation

Done via Google Sheet

Are there any open issues?

Are there any questions?

2. Agree on use case

3. Start collecting data

